CHOW'S LEMMA FOR SEPARATED FINITE-TYPE STACKS (GABBER)

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2458-2480 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 2482-2650, the complete author proof (167 body lines).

Editorial note. Navigation stubs: 19 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 2 of queue_stacks-stacks-more-morphisms.json (pb=7140, stmt 2458-2480, proof 2482-2650).

Editorial note. stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned. Named omission: the parenthetical 'details omitted' covers assumption verification only - the Gabber construction chain is in-span.

Editorial note. Statement carries a reference/slogan/history environment (patchenv -> patch-sd-envs.tex).

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'theorem-chow-finite-type' also exists in stacks-snapshot-20261011 — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original theorem lemma-make-section: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2231–2240. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 088N: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2650–2650. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:stacks-more-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original theorem 07SU: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2650–2650. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 0CN4: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2650–2650. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 0CPZ: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2650–2650. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 0CQ2: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2650–2650. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 0CPQ: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2650–2650. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 0CMK: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2650–2650. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 0CPV: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2650–2650. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 06Q9: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2650–2650. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 06R9: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2650–2650. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 06QI: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2650–2650. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 06QD: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2650–2650. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 0CPM: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2650–2650. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 0CPW: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2650–2650. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 0CML: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2650–2650. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 0CPT: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2650–2650. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 06RC: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2650–2650. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 050C: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2650–2650. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Theorem 0.1 (Chow's lemma)**.** *This is a result due to Ofer Gabber, see Let $f : \mathcal{X} \to Y$ be a morphism from an algebraic stack to an algebraic space. Assume*

(1) *Y is quasi-compact and quasi-separated,*
(2) *f is separated of finite type.*

Then there exists a commutative diagram

$$\begin{array}{ccccc} \mathcal{X} & \longleftarrow & X & \longrightarrow & \overline{X} \\ & \searrow & \downarrow & \swarrow & \\ & & Y & & \end{array}$$

where $X \to \mathcal{X}$ is proper surjective, $X \to \overline{X}$ is an open immersion, and $\overline{X} \to Y$ is proper morphism of algebraic spaces.

Proof. The rough idea is to use that $\mathcal{X}$ has a dense open which is a gerbe (Morphisms of Stacks, Proposition 06RC) and appeal to Lemma lemma-make-section. The reason this does not work is that the open may not be quasi-compact and one runs into technical problems. Thus we first do a (standard) reduction to the Noetherian case.

First we choose a closed immersion $\mathcal{X} \to \mathcal{X}'$ where $\mathcal{X}'$ is an algebraic stack separated and of finite type over Y. See Limits of Stacks, Lemma 0CQ2. Clearly it suffices to prove the theorem for $\mathcal{X}'$, hence we may assume $\mathcal{X} \to Y$ is separated and of finite presentation.

Assume $\mathcal{X} \to Y$ is separated and of finite presentation. By Limits of Spaces, Proposition 07SU we can write $Y = \lim Y_i$ as the directed limit of a system of Noetherian algebraic spaces with affine transition morphisms. By Limits of Stacks, Lemma 0CN4 there is an i and a morphism $\mathcal{X}_i \to Y_i$ of finite presentation from an algebraic stack to Y_i such that $\mathcal{X} = Y \times_{Y_i} \mathcal{X}_i$. After increasing i we may assume that $\mathcal{X}_i \to Y_i$ is separated, see Limits of Stacks, Lemma 0CPZ. Then it suffices to prove the theorem for $\mathcal{X}_i \to Y_i$. This reduces us to the case discussed in the next paragraph.

Assume Y is Noetherian. We may replace $\mathcal{X}$ by its reduction (Properties of Stacks, Definition 050C). This reduces us to the case discussed in the next paragraph.

Assume Y is Noetherian and $\mathcal{X}$ is reduced. Since $\mathcal{X} \to Y$ is separated and Y quasi-separated, we see that $\mathcal{X}$ is quasi-separated as an algebraic stack. Hence the inertia $\mathcal{I}_\mathcal{X} \to \mathcal{X}$ is quasi-compact. Thus by Morphisms of Stacks, Proposition 06RC there exists a dense open substack $\mathcal{V} \subset \mathcal{X}$ which is a gerbe. Let $\mathcal{V} \to V$ be the morphism which expresses $\mathcal{V}$ as a gerbe over the algebraic space V. See Morphisms of Stacks, Lemma 06QD for a construction of $\mathcal{V} \to V$. This construction in particular shows that the morphism $\mathcal{V} \to Y$ factors as $\mathcal{V} \to V \to Y$. Picture

$$\begin{array}{ccc} \mathcal{V} & \longrightarrow & \mathcal{X} \\ \downarrow & & \downarrow \\ V & \longrightarrow & Y \end{array}$$

Since the morphism $\mathcal{V} \to V$ is surjective, flat, and of finite presentation (Morphisms of Stacks, Lemma 06QI) and since $\mathcal{V} \to Y$ is locally of finite presentation, it follows that $V \to Y$ is locally of finite presentation (Morphisms of Stacks, Lemma 06Q9). Note that $\mathcal{V} \to V$ is a universal homeomorphism (Morphisms of Stacks, Lemma 06R9). Since $\mathcal{V}$ is quasi-compact (see Morphisms of Stacks, Lemma 0CPM) we see that V is quasi-compact. Finally, since $\mathcal{V} \to Y$ is separated the same is true for $V \to Y$ by Morphisms of Stacks, Lemma 0CPQ applied to $\mathcal{V} \to V \to Y$ (whose assumptions are satisfied as we've already seen).

All of the above means that the assumptions of Limits of Spaces, Lemma 088N apply to the morphism $V \to Y$. Thus we can find a dense open subspace $V' \subset V$ and an immersion $V' \to \mathbf{P}^n_Y$ over Y. Clearly we may replace V by V' and $\mathcal{V}$ by the inverse image of V' in $\mathcal{V}$ (recall that $|\mathcal{V}| = |V|$ as we've seen above). Thus we may assume we have a diagram

$$\begin{array}{ccccc} \mathcal{V} & & \longrightarrow & & \mathcal{X} \\ \downarrow & & & & \downarrow \\ V & \longrightarrow & \mathbf{P}^n_Y & \longrightarrow & Y \end{array}$$

where the arrow $V \to \mathbf{P}^n_Y$ is an immersion. Let $\mathcal{X}'$ be the scheme theoretic image of the morphism

$$j : \mathcal{V} \longrightarrow \mathbf{P}^n_Y \times_Y \mathcal{X}$$

and let Y' be the scheme theoretic image of the morphism $V \to \mathbf{P}^n_Y$. We obtain a commutative diagram

$$\begin{array}{ccccccc} \mathcal{V} & \longrightarrow & \mathcal{X}' & \longrightarrow & \mathbf{P}^n_Y \times_Y \mathcal{X} & \longrightarrow & \mathcal{X} \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ V & \longrightarrow & Y' & \longrightarrow & \mathbf{P}^n_Y & \longrightarrow & Y \end{array}$$

(See Morphisms of Stacks, Lemma 0CPV). We claim that $\mathcal{V} = V \times_{Y'} \mathcal{X}'$ and that Lemma lemma-make-section applies to the morphism $\mathcal{X}' \to Y'$ and the open subspace $V \subset Y'$. If the claim is true, then we obtain

$$\begin{array}{ccccc} \overline{X} & \longleftarrow & X & \xrightarrow{h} & \mathcal{X}' \\ & \searrow^{\overline{g}} & \downarrow{\scriptstyle g} & \swarrow_{f} & \\ & & Y' & & \end{array}$$

with $X \to \overline{X}$ an open immersion, $\overline{g}$ and h proper, and such that $|V|$ is contained in the image of $|g|$. Then the composition $X \to \mathcal{X}' \to \mathcal{X}$ is proper (as a composition of proper morphisms) and its image contains $|\mathcal{V}|$, hence this composition is surjective. As well, $\overline{X} \to Y' \to Y$ is proper as a composition of proper morphisms.

The last step is to prove the claim. Observe that $\mathcal{X}' \to Y'$ is separated and of finite type, that Y' is quasi-compact and quasi-separated, and that V is quasi-compact (we omit checking all the details completely). Next, we observe that $b : \mathcal{X}' \to \mathcal{X}$ is an isomorphism over $\mathcal{V}$ by Morphisms of Stacks, Lemma 0CPW. In particular $\mathcal{V}$ is identified with an open substack of $\mathcal{X}'$. The morphism j is quasi-compact (source is quasi-compact and target is quasi-separated), so formation of the scheme theoretic image of j commutes with flat base change by Morphisms of Stacks, Lemma 0CMK. In particular we see that $V \times_{Y'} \mathcal{X}'$ is the scheme theoretic image of $\mathcal{V} \to V \times_{Y'} \mathcal{X}'$. However, by Morphisms of Stacks, Lemma 0CPT the image of $|\mathcal{V}| \to |V \times_{Y'} \mathcal{X}'|$ is closed (use that $\mathcal{V} \to V$ is a universal homeomorphism as we've seen above and hence is universally closed). Also the image is dense (combine what we just said with Morphisms of Stacks, Lemma 0CML) we conclude $|\mathcal{V}| = |V \times_{Y'} \mathcal{X}'|$. Thus $\mathcal{V} \to V \times_{Y'} \mathcal{X}'$ is an isomorphism and the proof of the claim is complete. □